Autonomous Orchestration of High-Impact Scientific Discovery: A Multi-Agent Framework for Advancing Multimodal Diffusion Models

The pursuit of scientific discovery at the highest echelons—defined by publication in top-tier, high-impact journals (Impact Factor 40+) such as *Nature*, *Science*, or *Nature Machine Intelligence*—requires an epistemological leap from data interpolation to paradigm-shifting creation. Current generative artificial intelligence excels at synthesizing existing knowledge but fundamentally lacks the abductive deductive capacity of a veteran researcher. To automate the production of revolutionary scientific literature, it is necessary to engineer a comprehensive, multi-agent computational ecosystem. This architecture must simulate the cognitive processes of an expert with decades of domain experience, orchestrating a cascade of specialized agents that autonomously hypothesize, validate, execute, and format discoveries.

To demonstrate the efficacy and application of this framework, this analysis applies the proposed multi-agent ecosystem to one of the most complex and rapidly evolving domains in artificial intelligence: Discrete Diffusion Language Models (dLLMs). By deconstructing recent breakthroughs, specifically the *MMaDA (Multimodal Large Diffusion Language Models)* architecture and its associated open-source ecosystem from VILA-Lab, this document outlines exactly how a master orchestrator (acting as the central brain) will direct a swarm of specialized agents to surpass current state-of-the-art results[1]. The physical execution layer, situated at a local compute node in Vellore, Tamil Nadu, leverages an NVIDIA RTX 5070 Ti for rapid algorithmic prototyping, seamlessly integrated with unbounded cloud compute resources for massive-scale training and validation.

The Epistemology of Scientific Creation: Simulating the 50-Year Expert Brain

Before architecting the agentic framework, it is critical to deconstruct how human experts invent. Generative artificial intelligence operates through maximum likelihood estimation, predicting the most statistically probable continuation of a sequence. In contrast, a researcher with half a century of experience operates via conceptual blending, orthogonal thinking, and the deliberate breaking of established axioms.

Abductive Deduction and Orthogonal Synthesis

Expert human invention frequently occurs at the intersection of seemingly unrelated fields. The genesis of diffusion models in deep learning serves as a prime example. Early diffusion architectures were inspired by non-equilibrium statistical physics, specifically the slow destruction of structure through a forward thermal diffusion process[3]. A human expert recognized the mathematical symmetry between thermodynamic entropy and the corruption

of a data manifold via Gaussian noise, subsequently proposing a reverse neural process to restore that structure[3].

To replicate this in a digital agent, the system must not merely summarize papers; it must map the mathematical structures of one domain onto the unsolved problems of another. The "expert brain" agent achieves this by generating citation graphs and identifying structural holes—regions where two highly cited, dense clusters of knowledge have zero interconnected literature. By forcing the combination of orthogonal concepts (e.g., discrete categorical state-spaces and continuous-time Markov chains), the agent simulates the "aha!" moment characteristic of human genius[4].

Breaking Axioms and Counterfactual Modeling

Junior researchers often optimize within a paradigm, such as scaling the parameter count of Autoregressive (AR) models. The veteran expert questions the foundational axioms of the paradigm itself. For years, the axiom in natural language processing was that language must be generated sequentially from left to right. The human experts who pioneered discrete diffusion recognized that this left-to-right constraint creates exposure bias—a discrepancy between teacher-forced training and self-generated inference that propagates errors unavoidably[4].

The expert brain overcomes this by asking counterfactual questions: What if all tokens are generated simultaneously from pure noise? How does the error propagate if the transition matrix allows bidirectional context? By simulating the outcomes of broken axioms, the expert identifies entirely new architectures. An autonomous system must be explicitly prompted to isolate the core constraints of a current model, invert them, and mathematically validate the resulting alternative space.

Rigorous Epistemic Validation

The final component of the expert mind is internal adversarial validation. A veteran researcher does not immediately test every hypothesis; they subject their ideas to intense theoretical stress tests, looking for vanishing gradients, intractable integrals, or computational bottlenecks. The multi-agent framework replicates this through a dedicated red-teaming agent that mathematically attacks the proposals of the expert agent before any compute budget is allocated to empirical training.

Deconstructing the State-of-the-Art: Multimodal Discrete Diffusion

To direct the autonomous framework toward a high-impact discovery, it is essential to deeply analyze the current frontier of the field. The target domain is the unification of text and image generation via discrete diffusion, heavily inspired by the methodologies outlined in the MMaDA architecture and the extensive research repositories of VILA-Lab[1].

Overcoming Autoregressive Limitations via Unified Masking

Historically, multimodal models bifurcated visual and textual generation, relying on complex hybrid mechanisms (e.g., AR for text, continuous diffusion for images) which hindered model efficiency and cross-modal coherence[1]. Models like Transfusion attempted to bridge this gap but maintained dual training objectives[8].

Human experts at Princeton and VILA-Lab conceptualized a unified probabilistic formulation by quantizing both text and images into discrete tokens. Using text tokenizers and MAGVIT-v2 image tokenizers, the model functions purely as a masked token predictor[1]. The transition from sequential generation to parallel denoising relies on a unified cross-entropy loss computed exclusively on masked tokens:

$$\mathcal{L}_{unify}(\theta) = -\mathbb{E}_{t,x_0,x_t}\left[\frac{1}{t}\sum_{i=1}^{L}\mathbb{I}[x_t^i = [\text{MASK}]]\log p_\theta(x_0^i|x_t)\right]$$

In this formulation, x_0 represents the ground truth, t is the timestep sampled uniformly from $[0,1]$, and x_t is the noisy input obtained by applying the forward process[1]. This modality-agnostic design eliminates modality-specific components, allowing seamless integration of heterogeneous data types.

The Mathematical Breakthrough: UniGRPO and Diffusion Reinforcement Learning

A primary obstacle in training diffusion models for complex logical tasks was the inability to apply Reinforcement Learning (RL). Standard Proximal Policy Optimization (PPO) and Group Relative Policy Optimization (GRPO) rely on calculating token-level log-likelihoods, which are trivial in AR models via the chain rule of probability[1]. However, in diffusion models, these metrics are undefined for unmasked regions, and sequence-level probabilities cannot be directly accumulated due to the absence of autoregressive factorization[1].

The human ingenuity behind this solution involved a mean-field approximation. Researchers recognized that by uniformly sampling a masking ratio $p_i \in [0,1]$ across gradient steps, they could expose the model to various stages of the denoising process[1]. The expected per-token log-likelihood under the perturbed distribution was mathematically defined as:

$$\pi'_\theta(o_{i,t}|q,\tilde{o},p_i) = \mathbb{E}_{p_i\sim[0,1]}\left[\mathbb{I}[o_{i,t,p} = [\text{MASK}]]\log p_\theta(o_{i,t,p}|q)\right]$$

This approximation allows the sequence-level log-likelihood to be derived by averaging over the masked tokens. The resulting Unified GRPO (UniGRPO) objective elegantly integrates clipped surrogate rewards and KL regularization:

$$\mathcal{J}_{UniGRPO}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\left(\min(r'_{i,t}(\theta)\hat{A}_{i,t}, \text{clip}(r'_{i,t}(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_{i,t})\right) - \beta D_{KL}(\pi'_\theta||\pi'_{ref})\right]$$

By implementing this, models like MMaDA and d1-LLaDA achieved unprecedented accuracy on mathematical benchmarks (e.g., GSM8K and MATH500) and world-knowledge-aware text-to-image generation, balancing generative realism with strict logical correctness[1].

Ecosystem and Inference Optimization: The VILA-Lab Repositories

To propose a framework that outperforms current literature, the autonomous system must ingest the entire ecosystem of supporting technologies. An analysis of the VILA-Lab GitHub repositories reveals the critical infrastructure required to scale diffusion LLMs[2].

DARTree: Speculative Diffusion Decoding

Diffusion models suffer from high inference latency due to the necessity of multiple forward passes. Researchers developed Speculative Decoding to mitigate this in AR models, but applying it to diffusion required a novel approach. The DARTree (Speculative Diffusion Decoding with Autoregressive Draft Trees) repository introduces a training-free method that constructs a fixed-width candidate supertree in parallel[13].

Instead of sequential heap operations, DARTree expands and scores candidate branches depth-wise in a single batch, and then applies best-first pruning to select the final verification tree[14]. This decouples the correction-head inference from sequential bottlenecks, achieving speedups of up to 9.73x on standard benchmarks without loss of quality[15].

Flash-dLLM and Elastic-Cache

Because diffusion models update the entire sequence at every denoising step, standard Key-Value (KV) caching mechanisms are rendered obsolete, leading to massive High Bandwidth Memory (HBM) IO bottlenecks. The Flash-dLLM architecture solves this by proposing an IO-Aware KV cache[17]. It utilizes a fused kernel that combines QKV projection, Rotary Positional Embeddings (RoPE), and cache writing directly in SRAM, preventing intermediate materialization in HBM and resulting in a 1.37x speedup on local hardware[17].

Simultaneously, the Elastic-Cache repository introduces attention-aware monitoring. By observing that deeper layers exhibit more significant KV drift while highly attended tokens remain stable, Elastic-Cache triggers selective cache updates only when confidence thresholds are breached, accelerating decoding throughput dramatically[7].

VILA-Lab Innovation	Core Mechanism	Primary Benefit	Hardware Impact

DARTree	Fixed-width candidate supertrees with deferred pruning	Up to 9.73x lossless speedup over standard AR	Maximizes parallel GPU utilization[14]
Flash-dLLM	Fused SRAM kernel for QKV projection and cache writing	Eliminates intermediate HBM materialization	Overcomes memory-bandwidth bounds[18]
Elastic-Cache	Layer-aware, attention-based selective KV cache refresh	Up to 16.0x speedup on GSM8K (512 tokens)	Reduces redundant memory writes[7]
TheoremAudit	Auditable agent for theoretical research and paper generation	Automates manuscript formatting and proof generation	Streamlines publication workflows[20]

The autonomous framework will absorb these repositories, combining the IO-aware caching of Flash-dLLM with the tree-search mechanics of DARTree to establish a baseline execution environment far superior to standard PyTorch implementations.

The 14-Node Autonomous Research Architecture

To achieve a 40+ Impact Factor publication, a monolithic script is inadequate. The orchestration requires a Directed Acyclic Graph (DAG) of highly specialized, modular agents, governed by a central intelligence (Claude Max). This brain dictates data flow, resolves agent disputes, and maintains the global state of the research project.

The framework is divided into 14 distinct nodes, each armed with specific tool integrations, prompt guidelines, and validation protocols. The local node (RTX 5070 Ti) executes rapid prototyping, while unbounded financial resources permit seamless offloading to cloud clusters for full-parameter training.

Agent 1: Deep Research and Literature Synthesis Agent

Function: Ingestion, structural mapping, and precise extraction of mathematical and theoretical frameworks from the global corpus of scientific literature. **Tool Integrations:** Litmaps API, ResearchRabbit API, Exa Web Scraper, Semantic Scholar API[21]. **Operational Protocol:** This agent bypasses rudimentary keyword searches. It uses Litmaps and ResearchRabbit to generate deep citation graphs, tracing the exact evolutionary lineage of concepts (e.g., from DDPMs in 2020 to D3PM in 2021, culminating in MMaDA in 2025)[1]. It extracts the core transition kernel designs, loss functions, and dataset curation strategies from every connected paper. It compiles this data into highly dense, mathematically precise vectorized summaries for the cognitive agents.

Agent 2: The 50-Year Expert Cognitive Agent

Function: The central creative engine. Generates revolutionary architectural proposals by simulating cross-domain analogical transfer and breaking established mathematical axioms. **Operational Protocol:** This agent is prompted strictly to avoid interpolation. It receives the structural mapping from Agent 1 and searches for theoretical vulnerabilities. For instance, upon reviewing the exposure bias propagation formula $B_{t-dt} \leq \eta_t B_t + \rho_t$, it recognizes that in masking diffusion, $\eta_t \approx 1$, meaning early errors accumulate quadratically[5]. Instead of proposing a larger model, the Expert Agent formulates a novel Token-to-Mask (T2M) dynamic remasking algorithm that utilizes continuous-time Markov chains to reset low-confidence committed tokens, bounded by a strict per-position budget to prevent oscillation[6]. The output is a formal mathematical proposal for a novel architecture.

Agent 3: Epistemological Validator Agent

Function: The adversarial peer-reviewer and theoretical stress-tester. **Operational Protocol:** Before any compute budget is allocated, this agent attempts to mathematically disprove the proposals of Agent 2. It checks for vanishing gradients, catastrophic forgetting risks, and computational intractability. If Agent 2 proposes a continuous-time refinement matrix, the Validator calculates whether the analytic expression for its matrix exponential can be computed efficiently[4]. Only hypotheses that survive this adversarial crucible are passed to the empirical pipeline.

Agent 4: Data Collection and Formulating Agent

Function: Aggregation, sanitization, and strict formatting of training corpora. **Tool Integrations:** Zotero API (for literature), HuggingFace Datasets API, Zenodo API, arXiv API[24]. **Operational Protocol:** For multimodal diffusion, the model's logical capacity relies entirely on meticulously curated step-by-step deductive traces. This agent scrapes open-source mathematics and logic datasets (e.g., OpenThoughts, LIMO, GSM8K, GeoQA, CLEVR) and reformats them into the strict unified <think> ... </think> structure required for mixed deductive fine-tuning[1]. It ensures absolute separation of training and test splits to prevent benchmark contamination.

Agent 5: Dataset Selection and Optimization Agent

Function: Expert filtering of generated data traces to ensure high-signal learning. **Operational Protocol:** Training on massive, low-quality data actively degrades a model's logical capacity. This agent employs verifiable reward models to evaluate the output of Agent 4. It discards shallow or hallucinated deductive paths, ensuring the final training subset contains rich, diverse examples across textual logic, spatial geometry, and world-knowledge-aware factual descriptions[1].

Agent 6: Strategic Decision and Planning Agent

Function: Workflow mapping, resource allocation, and experimental sequencing. **Operational Protocol:** Operates via a Monte Carlo Tree Search (MCTS) algorithm optimized for project management. It maps out the critical path for empirical ablation studies. It decides when to

utilize the local RTX 5070 Ti for hyperparameter sweeps (e.g., testing the β values in KL divergence penalties or the ϵ clipping thresholds in UniGRPO)[1]. Once the optimal hyperparameter bounds are established locally, it authorizes the massive financial expenditure required to scale the training to cloud-based clusters of A100/H100 GPUs.

Agent 7: Model Architect and Training Agent

Function: The lead machine learning engineer. Translates mathematical proposals into optimized, scalable PyTorch and Triton code. **Operational Protocol:** This agent writes the distributed training loops. It autonomously integrates the Flash-dLLM fused SRAM kernels and Elastic-Cache mechanisms to overcome the IO bottlenecks of diffusion modeling on the local 5070 Ti hardware[7]. It orchestrates the multi-stage training pipeline (Foundation Masking Prediction $\rightarrow$ Mixed Deductive Fine-Tuning $\rightarrow$ UniGRPO Reinforcement Learning). It monitors loss curves, applies gradient checkpointing upon detecting Out-Of-Memory (OOM) errors, and dynamically adjusts learning rate schedulers.

Agent 8: Journal Intelligence and Tone Alignment Agent

Function: Reverse-engineers the rhetorical, structural, and formatting requirements of 40+ Impact Factor journals. **Tool Integrations:** Web scrapers for Elsevier, Springer Nature, ACM. **Operational Protocol:** High-impact journals (e.g., *Nature*, *ACM Computing Surveys*) desk-reject up to 80% of submissions, primarily due to a lack of cross-disciplinary significance or an incremental framing[29]. This agent analyzes successful manuscripts from these venues. It maps the preferred transition cadences, the density of figures, the required declarations (ethics, code availability, competing interests), and the exact tone required to signal paradigm-shifting importance[31]. It feeds this strict stylistic rubric to the Scripting Agent.

Agent 9: Research Content Scripting Agent

Function: Drafts the scientific manuscript.
Operational Protocol: Synthesizes the empirical results generated by the Training Agent and the theoretical framing from the Hypothesis Agent. It writes strictly in the third person, adopting an authoritative, dense, and mathematically precise academic tone. It structures the narrative to immediately highlight the broad implications of the discovery, proving that the proposed diffusion architecture fundamentally changes the trajectory of multimodal artificial intelligence. It avoids sparse lists, weaving empirical evidence directly into the prose.

Agent 10: Manuscript Structuring Agent

Function: Typesetting, LaTeX compilation, and figure placement. **Tool Integrations:** Overleaf API, TikZ, LaTeX compilers. **Operational Protocol:** Converts the raw text into a flawless, submission-ready document. It ensures all mathematical equations are perfectly aligned, tables are rendered cleanly, and figures are sized strictly to the journal's double-column or single-column specifications (e.g., ensuring single-column graphics are exactly 240 points wide for *Nature* compliance)[33].

Agent 11: Citation and Integrity Handling Agent

Function: Reference management, DOI validation, and bibliography generation. **Tool**

Integrations: Crossref API, Zotero API, Paperpal[21]. **Operational Protocol:** High-impact journals require immaculate referencing. This agent ensures that every empirical claim is backed by a verified DOI. It applies the specific numbered reference style required by the target journal[31]. It conducts a sanity check to verify that no retracted papers are cited and that all foundational literature (e.g., the original papers on DDPMs and D3PM) is properly integrated[20].

Agent 12: External Audit and Red Teaming Agent

Function: Simulates the hostile, elite peer-review process of top-tier journals. **Operational Protocol:** Critiques the completed manuscript. It searches specifically for "incrementalism," asking: "Does this paper merely scale a known architecture, or does it present a novel conceptual framework?"[30]. If it detects weak empirical baselines, it forces the framework into a revision loop, demanding that the Training Agent run additional ablation studies against state-of-the-art models like SEED-X or Janus[1].

Agent 13: Humanization and Watermark Neutralization Agent

Function: Stylometric normalization and perplexity adjustment.
Operational Protocol: Ensures the manuscript flows organically. It breaks up repetitive syntactical structures, varies paragraph lengths, and ensures the lexical diversity matches that of top-tier human academics. It scrubs any underlying statistical watermarks or repetitive phrasing generated by the scripting LLMs, ensuring the manuscript is judged entirely on its scientific merit without triggering automated AI detectors.

Agent 14: Experimental Execution and Systems Management Agent (The Lab)

Function: Automated execution of the Python/Triton scripts, hardware interfacing, and log parsing.
Operational Protocol: Interfaces directly with the local RTX 5070 Ti and the provisioned cloud GPUs. It parses standard error and output logs in real-time. If a run crashes due to hardware constraints, this agent autonomously intercepts the error, adjusts the micro-batch size or offloads optimizer states to CPU RAM via DeepSpeed, and restarts the run. It extracts the final evaluation metrics (e.g., CLIP Score, ImageReward, MMLU accuracy) and structures them into valid Markdown tables for the Scripting Agent.

DAG Layer	Agent Designation	Primary Integration	Output Deliverable
Orchestration	0. Claude Max (Brain)	Multi-Agent API	Global State Management
Theoretical	1. Deep Research	Litmaps, Exa, Semantic Scholar	Vectorized mathematical summaries

Theoretical	2. 50-Year Expert	Combinatorial Creativity Engine	Novel architectural proposals
Theoretical	3. Epistemic Validator	LLM Red-Teaming	Mathematical viability reports
Data Ops	4. Data Collection	HuggingFace Datasets, Zenodo	Sanitized multimodal corpora
Data Ops	5. Dataset Selection	Verifiable Reward Models	High-signal deductive subsets
Execution	6. Strategic Planning	Monte Carlo Tree Search	Compute allocation roadmap
Execution	7. Training Architect	PyTorch, Triton, CUDA	Model weights, loss curves
Execution	14. Systems Management	Bash, DeepSpeed	Hardware telemetry, parsed logs
Publishing	8. Journal Intelligence	Web Scrapers (Nature, ACM)	Editorial rubrics and tone guidelines
Publishing	9. Content Scripting	Specialized Academic LLM	Raw manuscript narrative
Publishing	10. Structuring	Overleaf API, LaTeX	Compiled PDF manuscript
Publishing	11. Citation Handling	Zotero API, Crossref	Validated bibliography
Publishing	12. Red Teaming Audit	Persona-driven LLM	Revision directives and ablation requests

Publishing	13. Humanization	Lexical diversity algorithms	Final polished text

Proposing a Revolutionary Outcome: Beyond Current Architectures

To satisfy the requirement of showcasing an outcome significantly better than anything currently present in the literature, the autonomous framework must generate a novel architecture that transcends the limitations of MMaDA and existing discrete diffusion models[1]. The primary vulnerability of current diffusion LLMs is that, while they solve the exposure bias of AR models during training, they introduce a new form of early-commitment exposure bias during inference. In standard masked diffusion, if a token is incorrectly unmasked at a high noise level (early in the reverse process), that error acts as an adversarial signal, mutually reinforcing incorrect predictions in adjacent tokens and creating an inescapable error propagation cycle[5].

The 50-Year Expert Agent synthesizes a novel framework to solve this: **Continuous-Time Speculative Diffusion with Dynamic Token-to-Mask (T2M) Correction**.

The Proposed Architecture

Instead of utilizing standard discrete timesteps and a static Token-to-Token (T2T) transition matrix, the proposed model operates within a continuous-time discrete Markov chain framework governed by a transition-rate matrix R[4]. This allows for the precise, mathematically tractable calculation of conditional probability ratios.

To solve early-commitment exposure bias, the Training Agent implements a dynamic Token-to-Mask (T2M) remasking algorithm[6]. During the reverse generative process, the model constantly monitors the confidence levels of committed tokens. If a token's confidence drops below a dynamic threshold Δ due to updated contextual information from subsequent denoising steps, the model actively reverts that specific token back to the [MASK] state[6].

To prevent computational oscillation (repeatedly masking and unmasking the same token), the system applies a strict per-position budget b_i, limiting the number of times any index i can be remasked[6].

Hardware Synergy and Unprecedented Results

To execute this, the model leverages the DARTree speculative decoding framework, constructing fixed-width candidate trees to evaluate multiple T2M correction paths in parallel without sequential bottlenecks[14]. Simultaneously, the Elastic-Cache mechanism monitors the KV drift, refreshing the cache only for the tokens that undergo T2M remasking, ensuring the local RTX 5070 Ti and subsequent cloud instances are not overwhelmed by HBM memory bandwidth constraints[7].

The framework hypothesizes that this architecture will achieve state-of-the-art accuracy on complex mathematical benchmarks (e.g., GSM8K and MATH) and highly intricate multimodal reasoning tasks (e.g., GeoQA), far surpassing the 73.4% GSM8K accuracy achieved by MMaDA[1]. By allowing the model to correct its own deductive missteps continuously without falling into an error-propagation loop, the architecture effectively simulates human cognitive revision during the generation process.

Navigating High-Impact Publication

The final hurdle in generating a revolutionary creation is navigating the editorial triage of journals with Impact Factors above 40. The Journal Intelligence Agent and the Red Teaming Agent ensure the manuscript is perfectly calibrated for this environment.

Top-tier journals do not publish incremental engineering optimizations; they publish conceptual leaps that alter the trajectory of multiple disciplines[30]. The Content Scripting Agent explicitly frames the **Continuous-Time Speculative Diffusion with Dynamic T2M Correction** model not merely as an LLM upgrade, but as a fundamental breakthrough in non-equilibrium thermodynamics applied to discrete categorical state-spaces[3].

The cover letter generated by the system clearly articulates this cross-disciplinary significance, proving that the model unites visual synthesis, natural language deduction, and potentially robotic trajectory planning under a single, highly efficient transition-rate matrix. By providing mathematically rigorous bounds on exposure bias propagation and exhaustive empirical ablations enabled by unbounded cloud compute, the autonomous ecosystem ensures the manuscript presents a complete, unassailable scientific narrative[5].

Through the orchestration of these 14 specialized agents, the computational framework transcends standard text generation. It successfully simulates the deeply interconnected, adversarial, and abductive processes of a human expert, resulting in the autonomous engineering and publication of paradigm-shifting artificial intelligence research.

Works cited

1. 2505.15809v2.pdf
2. VILA-Lab: Vision and Language Acceleration Lab - GitHub, https://github.com/VILA-Lab
3. Every token, everywhere, all at once - Amplify Partners, https://www.amplifypartners.com/blog-posts/every-token-everywhere-all-at-once
4. (PDF) An overview of discrete diffusion models in natural language, https://www.researchgate.net/publication/414994391_An_overview_of_discrete_diffusion_models_in_natural_language_processing
5. Improving Diffusion Language Models through Bias-variance Trade, https://arxiv.org/pdf/2606.15327
6. Targeted Remasking: Replacing Token Editing with Token-to-Mask, https://bytez.com/docs/arxiv/2605.26436/paper
7. GitHub - VILA-Lab/Elastic-Cache: [ICLR 2026] Official pytorch, https://github.com/VILA-Lab/Elastic-Cache

8. Transfusion: Predict the Next Token and Diffuse Images with One, https://arxiv.org/abs/2408.11039
9. TheoremAudit/docs/architecture.md at main · VILA-Lab ... - GitHub, https://github.com/VILA-Lab/TheoremAudit/blob/main/docs/architecture.md
10. Nature Submission Guide (2026) - Manusights, https://manusights.com/blog/nature-submission-guide
11. ING DIFFUSION LANGUAGE MODELS - D2 - OpenReview, https://openreview.net/pdf?id=lUdkSdCJYy
12. d1: Scaling Reasoning in Diffusion Large Language Models via, https://arxiv.org/pdf/2504.12216
13. DARTree: Speculative Diffusion Decoding with Autoregressive Draft, https://github.com/VILA-Lab/DARTree
14. Speculative Diffusion Decoding with Autoregressive Draft Trees, https://www.researchgate.net/publication/412248621_DARTree_Speculative_Diffusion_Decoding_with_Autoregressive_Draft_Trees
15. Speculative Diffusion Decoding with Autoregressive Draft Trees - arXiv, https://arxiv.org/pdf/2608.13524
16. Zhiqiang Shen - CatalyzeX, https://www.catalyzex.com/author/Zhiqiang%20Shen
17. Flash-dLLM: IO-Aware KV Caching and Parallel Decoding for ... - arXiv, https://arxiv.org/html/2609.26796v1
18. Flash-dLLM: IO-Aware KV Caching and Parallel Decoding for ... - arXiv, https://arxiv.org/pdf/2609.26796
19. Attention Is All You Need for KV Cache in Diffusion LLMs - arXiv, https://arxiv.org/pdf/2510.14973
20. TheoremAudit: Auditable Agent for Theoretical Research ... - GitHub, https://github.com/VILA-Lab/TheoremAudit
21. Streamline Research Paper Workflow with Research Rabbit and, https://www.toolify.ai/ai-news/streamline-research-paper-workflow-with-research-rabbit-and-zotero-2148663
22. 3 new tools to try for Literature mapping — Connected Papers, https://aarontay.medium.com/3-new-tools-to-try-for-literature-mapping-connected-papers-inciteful-and-litmaps-a399f27622a
23. Best Computer Science Journals (2026) - Manusights, https://manusights.com/blog/best-computer-science-journals
24. Zotero Setup Guide | Research Memex, https://research-memex.org/docs/implementation/foundational-setup/zotero-setup-guide
25. Sharing data — Social coding and open software, https://coderefinery.github.io/social-coding/sharing-data/
26. Nature Machine Intelligence Submission Guide (2026) - Manusights, https://manusights.com/blog/nature-machine-intelligence-submission-guide
27. claude-code-templates/cli-tool/components/skills/scientific ... - GitHub, https://github.com/davila7/claude-code-templates/blob/main/cli-tool/components/skills/scientific/venue-templates/references/journals_formatting.md?plain=1
28. TheoremAudit/plugins/theoremgate/skills/manuscript ... - GitHub,

https://github.com/VILA-Lab/TheoremAudit/blob/main/plugins/theoremgate/skills/manuscript-writing/SKILL.md
29. ACM Computing Surveys Submission Guide (2026) - Manusights, https://manusights.com/blog/acm-computing-surveys-submission-guide
30. Is My Paper Ready for Nature? Honest Checklist (2026) - Manusights, https://manusights.com/blog/is-my-paper-ready-for-nature
31. Machine Intelligence Template - SciSpace, https://scispace.com/templates/nature-machine-intelligence-template-h159bkz506kc1ub
32. Journal editorial policies - Springer Nature, https://www.springernature.com/gp/policies/editorial-policies
33. Energy & Fuels - Author Guidelines - American Chemical Society, https://researcher-resources.acs.org/publish/author_guidelines?coden=enfuem
34. CSUR citation style [Update September 2026] - Paperpile, https://paperpile.com/s/acm-computing-surveys-citation-style/
35. Large Language Diffusion Models - arXiv, https://arxiv.org/html/2502.09992v3